Research report

PA consistency machines

Comparison of the five U208 proposals and attempts to cross the 208-state threshold

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Status. The five supplied proposals reduce to two transition-graph classes, and both pass a fresh symbolic runtime replay. This investigation develops new source transformations and several independently checkable search methods. It has **not established a valid machine with fewer than 208 working states**. Rejected 207-state tables are not presented as solutions.

What the archive supplies

The complete LaTeX source and PDF; pinned upstream provenance; comparison witnesses and verification receipts; new arithmetic and source-level proofs; search implementations and scoped negative results; and instructions to reproduce the accepted checks. Executable experiments, ordinary mathematical proofs, and proof-assistant certificates are distinguished throughout.

Abstract

The requested endpoint is a two-symbol, one-tape machine M with fewer than 208 working states such that $\mathsf{PA} \vdash \neg\,\mathrm{Halt}(M, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathsf{PA})$. Five incoming U208 archives were examined at the same pinned repository revision. Their nine primary and alternative tables fall into exactly two rooted, action-preserving isomorphism classes. All five proposals implement the same local identification of historical states 172 and 211; their advertised savings cannot be added. Fresh verification covers 5,084 symbolic primitive contracts and every literal transition row for each graph class. The inherited consistency calibration was audited, including the difference between unary Theory U, the native $W + \mathsf{IR}$ calculus, and the separate positive theorem-stream proposal. New research includes a more general source associativity theorem, finite-window inductive context analysis, a direct 18-primitive negation constructor, fixed-role cover emitters, and a proof compiler with unnecessary successful-modus-ponens padding removed. Four new local constructions also yield eighteen pairwise non-isomorphic 208-state machines with short halting-equivalence proofs. None has yet yielded a verified sub-208 literal table. The negative results are explicitly limited to their search families; they are not state-count lower bounds.

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1 Problem, conventions, and evidential standard

The motivating K215 result is materially stronger than an informal statement of equiconsistency: its final theorem is proved in PA and binds a particular literal machine to a particular arithmetization of ZFC inconsistency [1]. Replacing its set theory, source program, or transition table requires new bridges at the affected levels. The K215 certificate does not automatically certify a U-based construction.

Here a working state is a state with two binary transition rows; the halt state is excluded from the count. A transition writes a bit, moves exactly one square left or right, and selects the next state. The tape is two-way infinite and initially all zero, with designated initial state 0. A smaller alphabet count, a nonblank input encoding, or a different machine convention would not answer the requested comparison.

There are four separate proof obligations:

1. A theory/calculus calibration, verified inside PA at the level of finite refutations, not merely an external equivalence of models.
2. Sound and complete enumeration of numerical derivations by the structured register program, including its startup invariant.
3. Correct implementation of every required primitive over arbitrary natural register values, complete tape frames, and arbitrary permitted exterior tape.
4. An exact state census and a binding of all those statements to the actual delivered transition table.

A finite blank-tape run, finite register tests, or a small source expression establishes none of these obligations by itself.

We use the following evidence labels. An *ordinary proof* is a written mathematical argument. *PA-formalizable* means its construction uses the specified finite syntactic transformations and induction available in PA; it does not assert that a new PA proof object has been generated and checked. A *symbolic runtime replay* applies explicit induction schemas and arbitrary-natural tape identities. A *finite screen* is used to reject candidates and is never promoted to an unbounded correctness theorem.

2 Pinned inputs and comparison of the five proposals

All incoming inputs were retrieved from ProveIt revision

`c086b5adaa6228009365768946d691dfc49a20bb`.

The Git blob identities were checked against the returned repository metadata, and the exact archive byte hashes were recorded before extraction. The input inventory is in `sources_manifest.json`; Appendix A lists archive hashes. No source package was edited in place.

For brevity, A and B denote rooted graph classes, preserving both written symbol and movement direction on every edge. They do not merely have the same state count or similar diagrams.

Across all nine selected primary and alternative tables, canonical rooted serialization gives exactly these two SHA-256 values:

Class A: `5d858b5d7528dd98c5272f83d308c782ae323a47fc394a73ad9efda1096a5284`

Class B: `f0c6098fabd68fce3fca0c6ac78a1b4e9dbb4ded60437b84d3e9d9ccda74f11d`

Archive shorthand	Primary	Relationship
ConPA	A	Retains historical state 172 as the merged representative.
Conditional Context	A	Retains 211 instead; supplies a class-B alternative.
Entry Context	A	Primary native and dense tables are byte-identical to Conditional Context.
History	A	Relabels the same graph; supplies a class-B alternative.
Research	B	Its Plain alternative is class A.

Table 1: Five packages, two literal transition-graph classes.

These are hashes of the comparison script's canonical serialization, not hashes of a particular archive's raw table bytes. The JSON witness records the corresponding state maps and raw-file hashes.

All tables have 208 working states and 416 rows. Both classes implement the same ordered 3,909-primitive source word and the same 5,083 ordinal control edges. Their source associations differ: the recorded maximum source depth is 42 for A and 41 for B. This distinction matters to the physical implementation because selector states are reused inside arithmetic code.

2.1 What the packaging differences mean

The Conditional Context title is not an admission that the local reduction depends on an unproved hypothesis. Its condition is a tape-context property proved by a complete predecessor analysis. The Entry Context and History presentations organize the same essential evidence differently. Selecting 172 rather than 211 as the retained state is a relabeling choice. Consequently there is no sequence of five independent one-state savings waiting to be composed.

For continued work, class A is a convenient main baseline because its lineage and state labels directly match the first ConPA archive. Class B is a useful independent source-layout alternative. This is a practical choice, not a mathematical preference between their consistency statements.

2.2 Fresh replay

After inspection, all five quick reproduction drivers were run with output outside the sealed input packages. Both distinct machines were then replayed through the Research package's full symbolic checker. The result was:

Quantity	Class A	Class B
Arbitrary-natural primitive contracts	5,084	5,084
Literal rows exercised symbolically	416/416	416/416
Symbolic inference applications	1,251,865	1,250,564
Exact blank startup transitions	48,140	48,140

The combined full replay took about 333 seconds in this environment. Dependency verification also checked 167 retained member hashes and the nested Theory U binding. The receipt is `analysis/comparison/full_research.json`; it should be read alongside the verifier's stated induction schemas, not as a new proof-assistant export.

3 The existing local 209-to-208 reduction

The reduction identifies historical states 172 and 211. Its point is that the surrounding tape distinguishes visits whose state names are being identified. The complete predecessor census establishes:

1. On every entry to old state 172, the two preceding tape cells are `10`.
2. On entry to old state 211 while reading zero, the immediately preceding cell is one.

These claims concern arbitrary initial tape runs from the specified initial state, not only the observed blank startup.

In the class-A representative, both transitions entering 211 from state 163 are redirected to 172, the zero row at 172 is altered, and state 211 is deleted. The key rows are

$$\begin{array}{lllllll} 163,0 & \mapsto & (0,L,211) & \longrightarrow & 163,0 & \mapsto & (0,L,172), \\ 163,1 & \mapsto & (1,R,211) & \longrightarrow & 163,1 & \mapsto & (1,R,172), \\ 172,0 & \mapsto & (0,L,67) & \longrightarrow & 172,0 & \mapsto & (0,L,217). \end{array}$$

The four old steps at $(172,0),(172,1),(211,0),(211,1)$ are simulated by new blocks of lengths $7,1,3,1$. Every block preserves the entire tape, has the correct state/head endpoint under the quotient, and has no interior halt. The other 414 old rows commute directly with the quotient.

Proposition 3.1 (Bounded positive stuttering)**.** *The old and new machines halt on exactly the same inputs. If an old run halts after T_{209} transitions, the corresponding new run has length*

$$T_{208} = T_{209} + 6N_{172,0} + 2N_{211,0} \leq 7T_{209},$$

where $N_{q,b}$ counts the indicated old steps before halting. The finite local argument and both halting implications formalize in PA.

Proof. Induct on old execution length, substituting the appropriate positive block at each step. This constructs the new boundary configurations and the stated time identity. For reflection, boundaries occur after positive blocks of length at most seven. An interior point cannot be a new halt. A new halting prefix therefore ends at a simulated old boundary; a bounded search through the new prefix recovers the corresponding old prefix. Finite tape windows and finite trajectories can be coded arithmetically, so the induction and bounded recovery require no consistency assumption. □

This local theorem is strong evidence for the shared reduction. It must still be composed with the theory, program, runtime, and startup bridges to obtain the final PA-consistency statement.

4 Eighteen new machines at the 208-state boundary

A further local synthesis found four useful merge-and-helper constructions. Each deletes one old state and adds one helper, so its count remains $208 - 1 + 1 = 208$. Different donor choices give nine graphs over each original class. An independent canonical comparison confirms eighteen pairwise non-isomorphic new graphs, all distinct from both original graphs. These are proved alternative starting points, not sub-208 solutions.

All labels in this section are the *History package's native labels*, which keep state 172 in the inherited fold. Let q be deleted, r retained, and let the new state be 500. Redirect all old targets q to r; replace the indicated row r/b by $(b,L,500)$; add the two displayed helper rows.

q	r	b	500/0	500/1	Lengths at q, r on b
127	124	0	$0, R, 8$	$1, R, 59$	3,3
190	160	1	$0, R, 86$	$1, R, 193$	3,3
178	168	0	$0, R, 154$	$1, R, 176$	3,3
301	88	0	$0, R, 122$	$1, R, 500$	5,3

In the first row, donor 59 can be replaced by 109 or 125. In the third, 154 can be replaced by 159 and 176 by 225, independently. This gives $3 + 1 + 4 + 1 = 9$ choices per baseline.

4.1 A one-step invariant suffices

The complete incoming census is particularly simple:

State	All incoming source/read pairs	Written predecessor bit
127	110/0	0
124	104/1	1
160	139/0	0
190	165/1	1
168	176/0, 178/0, 225/0	0
178	158/1	1
301	87/0	0
88	77/1	1

Every listed incoming edge moves right, and none of the states is initial. Thus every visit has the displayed preceding bit, on every initial tape. This proof does not depend on the larger finite-window analysis.

For the first three constructions, the new block moves left to read the predecessor tag, right to a donor, then right through a row matching the old exceptional action. Its final head is one square right of its old location, and the complete tape is unchanged. The other-symbol rows agree and use one step. For the last construction, starting at retained state 88 and head 0 while reading zero, the two blocks are

$$\begin{aligned} \text{old 301, preceding 0:} \quad & 88 \xrightarrow{0L} 500 \xrightarrow{0R} 122 \xrightarrow{0L} 137 \xrightarrow{0R} 30 \xrightarrow{0R} 45, \\ \text{old 88, preceding 1:} \quad & 88 \xrightarrow{0L} 500 \xrightarrow{1R} 500 \xrightarrow{0R} 122. \end{aligned}$$

Each touches only positions -1 and 0 before its endpoint, preserves their bits, and finishes at head 1 with exactly the old target. Every other old row commutes with the quotient.

Theorem 4.1 (New 208-state alternatives)**.** *Each of the eighteen delivered helper tables has 208 working states and halts exactly when its old U208 baseline halts, on every initial tape. The bounded local proof formalizes in PA. Composing with the inherited calibration gives the same ordinary PA-proof argument for* $\neg\,\mathrm{Halt}(M, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathsf{PA})$.

Proof. The incoming census supplies the required tag at every old boundary. The complete two-cell cases give positive blocks of length at most five, with no interior halt. Apply the same finite-run induction and halting reflection argument as for the original fold. For the three-step cases, $T_{\text{new}} = T_{\text{old}} + 2N_{q,b} + 2N_{r,b}$; for the last case it is $T_{\text{new}} = T_{\text{old}} + 4N_{301,0} + 2N_{88,0}$. □

The standard-library-only checker `code/verify_helper_machines.py` independently checks the literal patches, the complete incoming census, all positive two-cell blocks, the ordinary rows, the state counts, and the graph-class census. It passed for all eighteen tables. Each also passed 5,084

finite canonical primitive cases. The finite cases are additional regression evidence; the preceding local argument is the halting-equivalence proof.

The intended saving was to let two merges share one helper, producing $208 - 2 + 1 = 207$. All 4,832 tested oriented combinations per baseline failed the bounded screen. Fresh ordinary one-row fold searches on each of the eighteen new layouts also produced no survivor. These failures are restricted to those synthesis families.

5 Audit of Theory U and the native consistency calibration

5.1 Three interfaces that must be kept separate

The unary theory U is equality-free classical first-order logic over nonempty domains, in one binary relation E. Define

$$\mathrm{Sub}(x, a) \;\equiv\; \forall t\,(E(t, x) \to E(t, a)).$$

For every formula φ whose only possible free variable is x, U has the schema

$$\forall a\,\exists b\,\big[\forall x(E(x, b) \to \varphi(x)) \;\wedge\; \forall x\big(\mathrm{Sub}(x, a) \to (\varphi(x) \to E(x, b))\big)\big]\,. \tag{1}$$

The source variable a cannot occur free in φ. The output is a weak cover: every member passes the predicate, and every passing subset of the source is included. The axiom does not require the output's members to be exactly the passing subsets of the source.

The actual U208 source instead uses a Cantor-coded native calculus whose ordinary theory is $W + \mathsf{IR}$. Here W allows matrix parameters in the weak-cover schema, and IR is global irreflexivity. Its numerical truth alias is 0 and its refutation target is 1. The intended PA-verifiable consistency cycle is

$$\mathrm{Con}(\mathsf{PA}) \implies \mathrm{Con}(W + \mathsf{IR}) \implies \mathrm{Con}(U) \implies \mathrm{Con}(\mathsf{PA}). \tag{2}$$

The positive dyadic theorem stream in the original unary-theory archive is a third interface. It targets the atom $E(v_0, v_0)$ at code 0 and uses a negative seed concerning $D \equiv \forall v\,E(v, v)$. Its positive theory alone has a one-object total-relation model. Its calibrated assertion is nonderivability of the atomic target, equivalently consistency after the negative seed is included. Equating the consistency of that positive theory alone with $\mathrm{Con}(\mathsf{PA})$ would be an error. It is not the source compiled by the five current U208 packages.

5.2 Lower interpretation: why unary matrices suffice

The subtle direction is interpreting arithmetic in U. The supplied argument uses coextensionality, saturation, and numerical representatives rather than assuming global extensionality. Coextensionality is a congruence in the second argument of membership; its use in the first argument requires saturated containers. Numerical representatives are saturated, transitive objects whose member classes are linearly ordered, together with a first-order endpoint condition for represented initial segments.

The ordering of the proof is essential. Heredity, irreflexivity, greatest predecessors, and uniqueness of existing successors are proved first. Induction is established before totality of successor. Given a numerical counterexample n, one filters the failures below n using the fact that every member of n is a subset of n. If a failure lies inside n, take its least class and use its existing predecessor. If none lies inside n, apply the successor step from the greatest predecessor of n to n. Neither argument assumes that n already has a successor.

Arbitrary ambient parameters are internalized in a legal unary matrix:

$$Q(v) \;\equiv\; \forall p\,(\mathrm{BaseAndStep}_\theta(p) \to \theta(v,p)).$$

The base and successor step for Q follow from their corresponding components. This is an actual elimination of free parameters in the nonlogical instance, not a silent strengthening of the schema.

After comparison and least-counterexample results, bounded numerical comprehension uses the least failing numerical tuple. For a putatively missing filter at that tuple, use the unary matrix

$$\exists n\,(\mathrm{LeastFailure}(n) \wedge \chi(z,n)).$$

All parameters n are bound. A separate support hypothesis supplies an object a such that every selected z is a subset of a. The upper and lower clauses then yield the desired exact profile. Finite tuples are minimized by finitely many least-element arguments; tuple-coding objects are not assumed prematurely.

Successor, exact powers of numerical sources, common bounds, and bounded graph comprehension follow in that order. Ordered-pair profiles include entire coextensionality classes, so equal-coordinate and representative cases are covered. Histories are constructed with the old graph existentially bound inside the matrix and numerical parameters alone left free. This avoids using an illegal arbitrary graph parameter in U's schema. Addition, multiplication, and each translated PA induction instance then follow from these history constructions.

5.3 Upper interpretation and internal consistency equivalence

Interpret $E(x,a)$ as "bit x of a is one." PA proves that the translated $\mathrm{Sub}(x,a)$ implies $x \le a$. For each fixed arithmetic formula, induction constructs its finite binary characteristic code through a. That gives an exact filter, hence the weak-cover instance. Parameters are permitted in this PA induction, so the interpretation also validates W. PA proves irreflexivity because $2^a > a$.

This is a formula-by-formula induction argument and works in nonstandard PA models. It does not use a uniform arithmetical truth predicate for all formulas. For the internal statement, the required bridge consists of uniform proof templates: renaming, substitution, term-to-graph translation, relativization, and finite proof concatenation. Those syntactic operations are primitive recursive, with verification by formula and proof induction in PA. This supplies the stated ordinary argument for

$$\mathsf{PA} \vdash \mathrm{Con}(U) \leftrightarrow \mathrm{Con}(\mathsf{PA}).$$

The supplied work does not implement every arbitrary-proof translation or export a new end-to-end PA kernel certificate. The audit found no fatal mathematical defect in the proof, but this remaining formalization boundary must be reported accurately.

5.4 Native role emitter and its exceptional cases

Write I, R, Q for the native implication, membership, and universal quantifier constructors, and $F = 1$, $N(A) = I(A,F)$. The emitter uses matrix/output label a, source label b, and tested label c:

$$\begin{aligned} V &= I(R(a,b), Q(c, R(a,b))), & e &= R(c,a),\\ G &= Q(a, I(R(a,c), R(a,b))), & U_0 &= Q(c, I(e,a)),\\ L_0 &= Q(c, I(a, I(G,e))), & H &= Q(a, I(U_0, I(L_0, F))). \end{aligned}$$

If $c \neq a, b$, the emitted bundle is $N(I(V,H))$; otherwise it is truth code 0. Classically the enabled bundle is $V \wedge \neg H$.

For positive code a, every effective variable label in its decoded formula is below a; zero is a closed truth alias. Thus the output binder is fresh. For $a \neq b$, G is the intended subset condition. For $a = b$, its binder captures the source and gives

$$\forall v_a \, [E(v_a, v_c) \to E(v_a, v_a)],$$

which is $\mathrm{Emp}(v_c)$ under IR. W supplies an empty source from a false matrix and then supplies precisely the required cover at that source. The codes $a = 0$ and $a = 1$ are treated explicitly as closed truth and falsity under IR. The gate is still needed for the separate atomic-vacuity component V.

For completeness, rename source and test to 0 and 1, double-negate the desired matrix to a code $n > 1$, and choose $(a, b, c) = (n, 0, 1)$. Project the cover, remove the double negation, and restore names. The inherited logical rows and equality-indiscernibility translation remain necessary premises. In particular, a supplied Boolean counterinterpretation rules out simply deleting the dominated propositional row.

5.5 Why the startup invariant cannot be omitted

At exactly 48,140 blank-tape transitions, the selected class-A table reaches native dispatcher 84 at head coordinate -2, source ordinal 136, with logical registers

$$(1, 0, 13, 0, 0, 1, 0, 0, 1).$$

The header is at 54, there are 43 spare blocks of value one, and the complete tape has 120 one-cells. These are exact finite data, not an extrapolation. The source address is `001100110101010`.

The source then enters an interior axiom suffix. Its incoming cache 1 is an arithmetic parameter, not an already proved falsity, and its numerical stack tail 4 is opaque. The retained suffix proof constructs nine explicit theorem heads above that tail. A separate ghost stack length is required because Cantor pairing has the zero alias $P(0, 0) = 0$. The scheduler's maximum initial conservative debt is two, leaving a permanent margin of seven. This unbounded credit argument, together with the suffix contract, prevents the opaque tail from being exposed. The finite startup trace alone does not establish this fact.

5.6 Simple theory shortenings that fail

Several tempting modifications have small countermodels, uniformly verifiable in PA:

- Deleting the lower clause permits a one-object empty relation.
- Deleting the upper clause permits a two-object directed cycle, even with irreflexivity. Each profile is a singleton and $\mathrm{Sub}(x, a)$ reduces to $x = a$.
- Replacing the subset guard by $E(x, a)$ again permits the one-object empty relation.
- Removing the guard while retaining both clauses makes the matrix $\neg E(x, x)$ yield Russell's contradiction.

These are counterexamples to the specified edits, not a lower bound on possible PA-equiconsistent schemas.

6 New source-level transformations

6.1 Directional associativity

The structured source language uses binary sequencing trees. A decrement's zero branch and a forward break select zero-edge ancestors; backward continues select one-edge ancestors. Let $f(X)$ and $b(X)$ be the respective escaping control-depth bounds. For a leaf they are

	f	b
CIr, CH	0	0
CDr	1	0
CBk	k	0
CCk	0	k

and at a sequence (X, Y) they satisfy

$$f(X,Y) = \max(f(X) - 1, f(Y), 0), \qquad b(X,Y) = \max(b(X), b(Y) - 1, 0).$$

Theorem 6.1 (Directional associativity)**.** *The trees $((A, B), C)$ and $(A, (B, C))$ have the same ordered primitive word and ordinal control graph whenever $f(A) = 0$ and $b(C) = 0$. No condition is needed on the middle factor B. Halt occurrences are allowed.*

Proof. Within A, forward controls do not escape. The prefix changes from 00 to 0, which changes no one-edge ancestor, so backward controls retain their targets. Within B, compare prefixes 01 and 10. The first escaping forward control sees the end of ABC in either tree: the old root and the new BC interval have the same right endpoint. The first escaping backward control sees the start of ABC: old AB and the new root have the same left endpoint. Every later relevant ancestor is outside the changed region and agrees. Within C, backward controls do not escape. Its prefix changes from 1 to 11, which changes no zero-edge ancestor, so forward targets agree. Internal targets are unchanged, as are ordinary successors and positive decrement branches in the identical primitive word. Halting has no target to change. □

The supplied builders used the stronger requirement that all three factors be completely closed and halt-free. On class A, the new theorem increases the initially eligible single-entry pure-selector rotations from 51 to 58. All 58 were checked against two independent CFG resolvers, including all 5,084 arithmetic target checks. Only one of the seven newly admitted literal layouts passed the finite physical screen; it did not save a state.

This last qualification matters: a source associativity theorem is not a theorem about the physical table. A selector state may also be called from an arithmetic routine, where its old row still has another required role.

6.2 Direct negation without a stack round trip

Let the native pairing be

$$P(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \qquad I(a,b) = 4P(a,b).$$

The original negation constructor pushes a literal one and uses generic implication. It consumes 47 source primitives and performs an unnecessary push/pop round trip. The following identity gives a shorter construction:

$$P(n,1) = P(0,n) + 2. \tag{3}$$

It is an elementary polynomial identity, valid also at $n = 0$.

Use the inherited pair routine with effect

$$(X, T, Y) \longmapsto (0, 0, Y + P(X, T))$$

and the inherited quadruple-transfer routine $Y \mapsto 4T$, which clears Y when T is initially zero. Start with $T = n$, $X = Y = 0$ and perform

$$Y \mathrel{+}= 1; \quad Y \mathrel{+}= 1; \quad \text{Pair}; \quad \text{QuadrupleTransfer}\,.$$

The resulting top is $4(P(0, n) + 2) = I(n, 1) = N(n)$. The stack and selectors are untouched, and both scratch registers end at zero. The routines require $2 + 10 + 6 = 18$ primitives. An alternative 20-primitive version first transfers T to X, sets T to one, then pairs and quadruples.

Four exact old 47-primitive subtrees can initially be replaced after the protected startup prefix. Two further occurrences are hidden by source association but are still closed 47-primitive intervals. Replacing all six with the 18-primitive routine saves $6(47 - 18) = 174$ occurrences, giving 3,735 primitives. The first 137 source rows are unchanged, all exterior primitive effects and control edges are preserved, and the logical axiom family is unchanged. The best initial literal substitutions have 198 distinct source DAG nodes, versus 197 in the baseline. Exploratory allocation yielded a 211-state four-occurrence implementation and a 212-state six-occurrence implementation, both passing their finite primitive screens. Thus this is a real arithmetic simplification but, by itself, not yet a state reduction.

6.3 Fixing cover roles while preserving the logical interface

The completeness proof already uses source label 0 and tested label 1. This suggests replacing those roles by constants throughout the cover, while retaining the original atomic-vacuity term and its original gate. With matrix/output selector n, define

$$\begin{aligned} e_n &= R(1, n), & g_n &= Q(n, I(R(n, 1), R(n, 0))), \\ u_n &= Q(1, I(e_n, n)), & \ell_n &= Q(1, I(n, I(g_n, e_n))), \\ h_n &= Q(n, I(u_n, I(\ell_n, F))). \end{aligned}$$

For $n > 1$ this is the intended weak-cover shell. The small cases must be checked, since the original vacuity gate does not necessarily exclude them. For $n = 0$, the inner guard binder also captures the fixed source; under irreflexivity it means emptiness, and W supplies the required cover at an empty source. For $n = 1$, the matrix is false under irreflexivity and the output/test collision makes the shell an irreflexivity instance. Thus the enabled fixed-role bundle remains sound in $W + \mathsf{IR}$.

Every original atomic-vacuity instance is still projected from its enabled bundle. For any desired W instance, padding to a matrix code $n > 1$ and choosing $(a, b, c) = (n, 0, 1)$ recovers its cover. The same proof templates give the original consistency calibration.

Replacing five test-label loads by literal-one loads and one source-label load by a literal-zero load saves 37 primitives, giving 3,872 occurrences. The first 137 rows and the exterior interface remain unchanged. Exchanging the upper and lower clauses inside the curried negation is classically equivalent; exchanging the two lower-clause antecedents is also legitimate. Using both exchanges improved the initial optimized source DAG from 206 to 203 nodes. This is still larger than the original 197-node source.

It is unsafe simply to reverse the constants without a repair: test 0, source 1 fails at matrix code 0. Clearing the entire enabled bundle there would also remove required atomic-vacuity instances. A sound repair sets the cover's negative shell to F before combining it with V; then the bundle becomes double-negated V. The explored variants record this distinction.

6.4 Removing successful-modus-ponens padding

The current successful MP command leaves an additional zero at the head of the proof stack. In work-register notation (T, W, D, H) its successful postcondition is

$$(z, P(0, w), z, 0).$$

That extra zero is a theorem but carries no information needed by the enumerator. A revised success branch can return $(z, w, z, 0)$ instead, leaving failure unchanged. This deletes the closed PUSH0 interval $[3847, 3866)$, or 19 primitive occurrences.

Soundness is preserved because no new top, cache value, or stack entry is introduced. The conservative stack-credit proof assigns MP weight zero already; it never relied on the additional head. Completeness must use the *final three-command compiler*, not an earlier four-command NIP/DROP presentation. Write $M = I(A, Y)$ and retain the established PUSH, DISCARD, and testing-cleanup H blocks. The MP-node compiler becomes

$$\mathrm{DISCARD}; \mathrm{PUSH}; B_M; \mathrm{PUSH}; B_A; \mathrm{PUSH}; S; S; \mathrm{MP}'(Y); H.$$

The initial PUSH reserves a zero. Immediately before the two S commands the tuple is $(0, [A; M; 0; w], 0, 0)$. The two transfers give $(A, [0; w], 0, M)$; the successful new MP gives $(Y, [0; w], Y, 0)$; and H removes the reserved zero, tests Y, and returns $(Y, w, 0, 0)$ or halts when $Y = 1$. The other uses of MP inside PUSH and DISCARD are forced failures, so their contracts are unchanged. The zero exception $I(0, 0) = 0$ is handled by leaving $J(0)$ empty. Induction on finite proofs therefore supplies the required strong Build contract, cofinal enumeration, and halting of a compiled refutation.

The resulting source has 3,890 primitives and 197 distinct DAG nodes. An exploratory literal allocation produced a 209-state implementation passing 10,118 finite primitive cases, with all 418 rows visited. This is not a sub-208 result and is not supplied as an accepted replacement for U208.

6.5 Direct selection and a checked zero invariant

The old S command halts exactly when the cache $h = 1$; otherwise it installs h as the current formula and pops the stack into the cache. Its 35-primitive implementation can be replaced by a 25-primitive routine: clear T; attempt one cache decrement; if it fails, proceed to the pop; otherwise increment T, attempt a second decrement, and halt if that fails. If the second succeeds, increment T again and transfer the remaining cache to T before the pop. For $h = 0$ this leaves $T = 0$; for $h = 1$ it halts; for $h \geq 2$ the transfer loop has invariant $T + H = h$ and decreases H to zero. The unchanged 14-primitive pop then has its original all-natural precondition. A 26-primitive version first copies the cache and performs the two-decrement test on scratch. Their source DAG counts are 201 and 199, respectively, so neither count constitutes a state improvement.

An independent product-domain analysis also proves a local zero fact useful to MP. At every source ordinal it records which registers are zero, positive, or unrestricted. The finite certificate checks 5,080 abstract CFG edges and has 3,907 nonempty ordinal domains. It proves T is zero at ordinal 3847 and excludes the positive body at 3848–3849. It includes both the canonical source seed and the actual interior bootstrap: at ordinal 136 its only requirements are $N > 0$ and scratch $Y = 0$, satisfied by the calibrated register vector. Closure of all abstract edges makes this an induction over arbitrary execution length. Deleting only that dead initial transfer gives 3,906 primitives but 198 source DAG nodes.

7 Context analysis and the negative searches

7.1 Finite-window inductive contexts

Finite histories constrain the tape near a state but forget information older than the chosen history length. A complementary analysis computes a least fixed point on states and radius-seven tape windows. The abstract window contains 15 bits centered on the current head. Initially, every window at state 0 is admitted, corresponding to arbitrary initial tapes. Each transition updates the scanned bit, shifts the window by one cell, and admits both possibilities for the newly exposed exterior bit.

Proposition 7.1 (Soundness of window closure)**.** *Any set of abstract state/window pairs containing every initial window and closed under both exposed-bit successors contains the abstraction of every concrete run from state 0.*

Proof. Induct on run length. The concrete new exterior bit is one of the two abstract choices. All other new window cells are determined by the old window and the literal transition, so the concrete successor belongs to the closed set. □

For class A the computed fixed point contains 961,765 abstract configurations, compressed into 2,692 exact Boolean cylinders. Class B has 2,698 cylinders. No working state is restricted to reading just one symbol. These are complementary unbounded-history invariants within the chosen window vocabulary. A long finite history can retain correlations involving a cell that a sliding window has forgotten, so there is no blanket dominance claim. The builder and its data are retained; independent transition-closure verification is the appropriate certificate boundary.

7.2 Search families and what their failures establish

Table 2 records selected completed searches. Counts describe the actual restricted families, not all possible smaller machines. Random or finite canonical checks are rejection filters; any survivor would still need an exact local simulation or a fresh symbolic runtime proof.

Table 2: Selected negative searches.

Family	Scope	Result
One-row shared-row contextual folds	369,152 mutations per graph class	No accepted fold with the computed radius-seven contexts.
General one-row representative helper	42,642 oriented pairs	No finite-screen survivor.
Two-row representative/helper changes	235,136,640 mutations over 174 shared-row pairs	No survivor in the completed broad screen.
Equal-tape head-offset quotients	37,053 choices, offsets ± 2	No accepted simulation.
Offset plus one retained-row mutation	61,656,192 choices	No finite-screen survivor.
Offset with predecessor-bit complement	61,656,192 choices	No finite-screen survivor.
Closed source rotations through depth 3	24,321 distinct tables	50 exact 207-state contractions; all fail finite physical validation.

Family	Scope	Result
Combined individually passing rotations	767 compatible nonempty subsets	No shared-row contextual-fold survivor.
Directional zero-run beam	19,136 tables, depth at most 20	Six-zero leaves reduced from 33 to 17; maximum run remains six.
Affine source summaries	25 total affine transformations	No universal equal-function decoder donor.
Uniform increment-run reorderings	24 register orderings in each class	Baseline DAG count 197 remains minimal in this family.

The head-offset screens adjust incoming movements and sometimes written symbols, so they explore a different quotient representation from ordinary same-head state merging. The representative/helper family allows a second modified row but requires every original use of that helper to remain valid. This preservation requirement is substantial; dropping it silently would make many apparent reductions unsound.

Source rearrangement also has a specific repeated obstruction. Within maximal single-entry pure-selector islands with closed atoms, the only repeated multi-atom factor exposed by arbitrary reassociation is the factor represented by states 12 and 150 at roots 52 and 155. It leads back to the same invalid 85/154 suffix identification. This explains repeated discovery of the same false saving within that family; it is not a general minimality theorem.

7.3 Counterexamples that prevented false positive results

A dispatcher visit is not necessarily a primitive boundary. Cutting execution every time native state 84 is visited produced an apparently useful invariant and the proposed deletion of state 14. That cut is invalid: a primitive may re-enter dispatcher 84 on a temporary program counter, before its canonical contract is complete. The finite checker rejects the candidate at the decrement instruction with address `0000100`. Those exploratory files are explicitly classified as rejected and are not certificates.

Matching scanners on short words is insufficient. A two-sided joining search temporarily retained a $52 \to 155$ alias. Exact branch analysis reaches state 32 on one side and 48 on the other. Both scan ones to the right, but their zero transitions count different numbers of separators. A longer unary block exposes the mismatch. Other scanner aliases similarly fail on long zero separators.

A smaller decoder graph may break arithmetic. Fifty 207-state contractions obtained after source rotations fail finite primitive tests even though their source graphs are correct. Selector states are reused as arithmetic continuation code. An independent one-row repair search over 86,008 choices did not repair the relevant failure.

A shorter zero detector needs a global invariant. Deleting state 34 from the origin detector fails an actual decrement case. The leftward search encounters temporary phase markers as well as final source addresses. Neither a shorter header constant nor an observed short source address proves that a six-zero detector is sound. The explored reassociations did not eliminate every six-zero source run in any event.

8 What is established, and what remains open

The comparison establishes two original U208 graphs, a single shared historical fold, and successful fresh symbolic replay of both. The eighteen new helper layouts have independently checked local halting-equivalence proofs and remain at 208 states. The theory audit supports the ordinary PA-formalizability claim while keeping its missing end-to-end proof-assistant export explicit. The new directional associativity theorem, the direct negation identity, and the revised MP proof-stack protocol are mathematical contributions with precise interfaces. Fixed-role emitters preserve the necessary native logical calibration, including exceptional small codes and atomic vacuity.

The requested state-count breakthrough remains open in this investigation. The central implementation obstacle is shared code: minimizing primitive occurrences, distinct source nodes, and literal machine states are three different optimization problems. A source routine may become shorter while losing a physical overlap that previously saved a state. Conversely, a slightly longer routine might enable better state sharing.

A defensible next candidate should therefore be accompanied by its literal table at discovery time. The decisive tests are a complete state census, the exact protected blank startup, arbitrary-natural primitive simulation over complete frames, and the applicable source/calculus proof. For a local quotient, a complete finite context invariant and positive bounded local simulation may replace the full runtime replay. No amount of additional blank-tape running substitutes for those arguments.

A Input hashes

U208_ConPA_2026-10-29.zip
`0662a968902f231ac2483bbd424f80b8f86e6fe1e04bbe97b6f25c634aa532f9`

U208_ConPA_Conditional_Context_2026-10-29.zip
`6f23aaca025984a9a3561c7cf352cbde069096713269d8ace9b8c0e305f3b251`

U208_ConPA_Entry_Context_2026-10-29.zip
`00b0a699606a654bcc8b2079932ec3c6c7303b0adc019e548db2b12d62eb4868`

U208_ConPA_History_2026-10-29.zip
`2479f562ba30b68bcabcc9ab481cdcb20da2a74476bff132a99382a6669a88aa`

u208_research_2026-10-29.zip
`33666892d7e729084edb5aa6e2b5e07d0117adbb63b0f46baff5bbbdf1da3d09`

pa_unary_power_cover.zip
`0fcb1117a5353766a3366bab04e56b4efbd0607833cdc73ffb34b67e6808b0ff`

B Reproduction and artifact map

`README.txt` is the entry point for the archive. The complete ordered file inventory and SHA-256 values are recorded separately. The supplied reproduction driver checks the pinned upstream archives, exact rooted graph comparison, the new source changes and their structural assertions, and selected counterexamples. Full symbolic replay is an explicit additional mode because it takes several minutes.

The most useful analysis records are:

- `analysis/candidate_comparison.md` and `analysis/comparison/current_comparison.json`: raw hashes, graph witnesses, and package-by-package evidence.

- `analysis/theory_audit.md` and `analysis/theory_lineage/`: detailed mathematical audit and exact inherited proof excerpts.
- `analysis/directional_associativity.md`: the new source theorem and the exact limits of its physical implications.
- `analysis/runtime_redesign/`: arithmetic and fixed-role emitter experiments, source trees, interfaces, and receipts.
- `analysis/reduction_search/notes.md`: scoped local searches, window invariants, rejected approaches, and MP protocol work.

All source-level variants are labeled experimental unless a separate literal machine theorem is explicitly supplied. Names such as `candidate` in a search directory denote search inputs, not accepted mathematical claims.

References

[1] Elliot Glazer and collaborators. *k215-pa: PA formalization for K215*. https://github.com/elliotglazer/k215-pa.

[2] Elliot Glazer and collaborators. *k230-rocq*, especially "The theory K." https://github.com/elliotglazer/k230-rocq/blob/main/README.md#the-theory-k.

[3] `jeff87654` and collaborators. *busy-beaver, ZF/m254*, branch `zf-254`. https://github.com/jeff87654/busy-beaver/tree/zf-254/ZF/m254.

[4] Vladimir Reshetnikov and collaborators. *ProveIt: Busy Beaver research and formal verification*. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Computability/BusyBeaver/Research/ZFC/k225-formal/verification/k215.

[5] ProveIt incoming submissions, revision `c086b5adaa6228009365768946d691dfc49a20bb`. https://github.com/VladimirReshetnikov/ProveIt/tree/c086b5adaa6228009365768946d691dfc49a20bb/docs/incoming.

[6] *pa_unary_power_cover.zip*, exact input identified in Appendix A; proof source and supporting artifacts in the `pa_unary_power_cover` package.

[7] The five U208 archives listed in Table 1, with exact input hashes in Appendix A. Local theorem proofs, runtime induction engines, and retained attribution notices are preserved in their respective packages.